

- 4 Holstein Friesian cattle are a breed of cattle used by dairy farmers in many countries of the world for the high milk yield of their cows.

Fig. 4.1 shows Holstein Friesian cattle.



Fig. 4.1

Milk yield in Holstein Friesian cattle is affected by heat stress. Heat stress occurs when homeostatic mechanisms are **not** enough to keep the body temperature down to normal levels.

One of the factors that contributes to heat stress is air temperature.

Fig. 4.2 shows:

- the mean daily air temperature in Central Europe
- the mean monthly milk yield per cow of Holstein Friesian cattle in Central Europe.

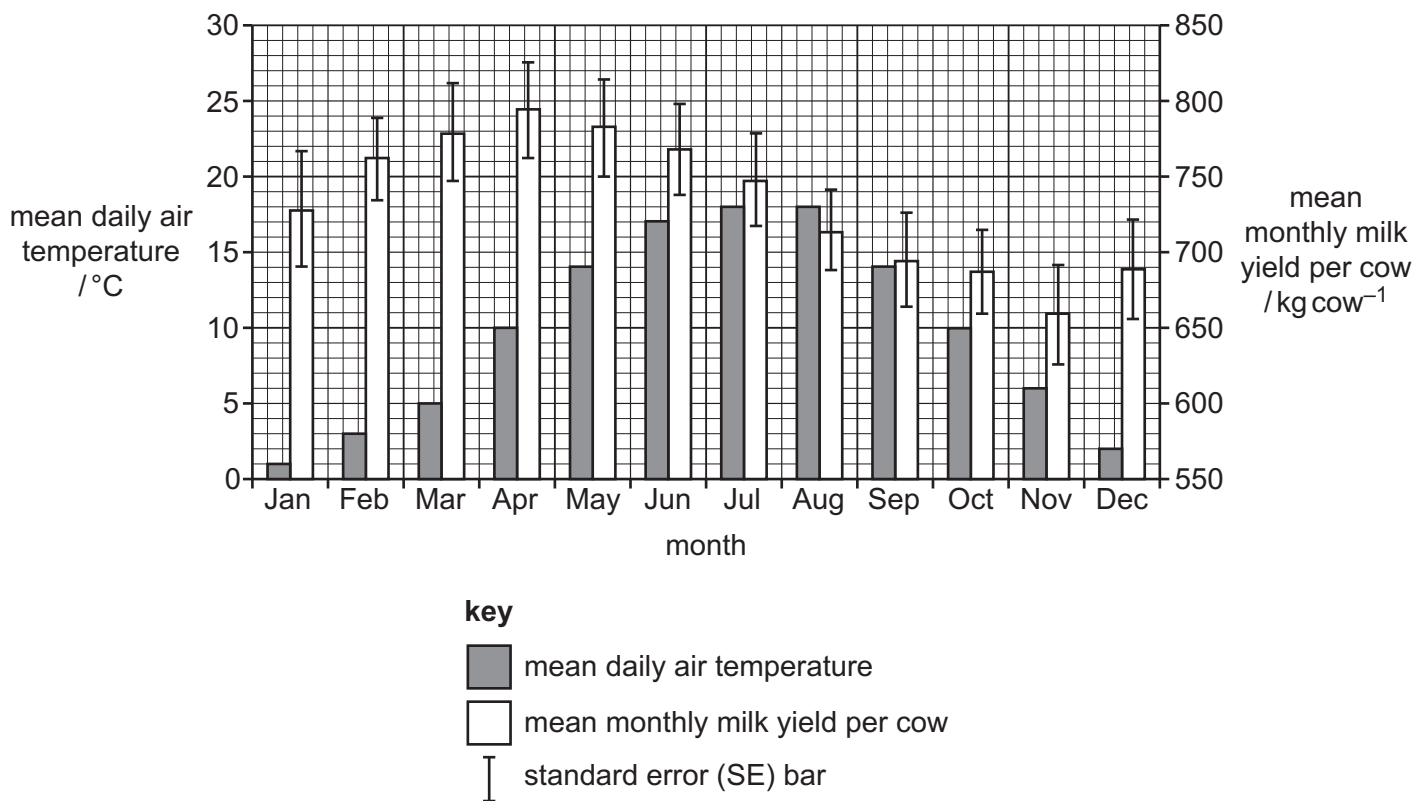


Fig. 4.2

(a) With reference to Fig. 4.2, describe the trends in air temperature and milk yield from April to August.

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(b) Many dairy farmers in tropical regions use cattle breeds that are tolerant to heat stress (heat-tolerant cattle). These heat-tolerant cattle:

- can tolerate higher air temperatures than Holstein Friesian cattle before heat stress occurs
- have milder symptoms of heat stress than Holstein Friesian cattle for the same high air temperatures.

Where heat stress does **not** occur, heat-tolerant cattle produce a lower milk yield than Holstein Friesian cattle under the same conditions.

Scientists compared DNA sequences of Holstein Friesian cattle and heat-tolerant cattle for a number of genes known to have an effect on body temperature.

Twenty genes were found that had alleles associated **only** with heat-tolerant cattle.

With reference to the information provided, including the data in Fig. 4.2:

- state the type (pattern) of phenotypic variation shown by milk yield in cattle
- identify factors that cause phenotypic variation in milk yield in cattle.

In each case, give a reason for your choice.

type (pattern) of phenotypic variation and reason for choice

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factors that cause phenotypic variation and reason for each choice

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- (d)** The *SLICK* allele differs from the recessive allele by a single nucleotide deletion. This results in a frameshift mutation and introduces a premature stop codon in the *PRLR* gene.

Scientists can use gene editing to replicate this mutation in Holstein Friesian cattle. This provides a way to introduce the *SLICK* allele into Holstein Friesian cattle without selective breeding.

Compare gene editing and selective breeding for introducing the *SLICK* allele into Holstein Friesian cattle.

Include similarities **and** differences in your answer.

[6]

[Total: 14]